

Complement Flow Synthesis with In-Line Purification — Using Flash Chromatography

A smooth approach for continuous isolation of flow synthesis products with the Advion Interchim Scientific puriFlash® 5.250 prep LC System

A key problem in flow chemistry is effectively isolating the desired products from residual starting materials, catalysts and by-products. This complicates the purity and yield of the desired product. Isolation methods involve manual intervention, at large, requiring multiple steps. This process is tedious and quite unproductive. Recently, advancements in preparative liquid chromatography have significantly improved the isolation of flow synthesis products. This application note describes a novel method to perform the continuous isolation of flow synthesis products from residual starting materials, catalysts or by-products to expedite chemical discovery. A promising new approach is highlighted here, featuring the Advion Interchim Scientific puriFlash® 5.250 preparative LC system.

The first results of our cooperation with the VilelaLAB and Continuum Flow Lab at Heriot-Watt University, Edinburgh, on this topic are outlined here.

Keywords: Isolation, Flow Synthesis, Preparative LC, Liquid Chromatography, Flash Chromatography, Advion, puriFlash® 5.250

Introduction

The continuous isolation of flow synthesis products from residual starting materials, catalysts or by-products is desirable to expedite chemical discovery. A promising new approach to achieve this goal is to distribute the effluent from an optimized flow reactor setup into two alternating flash chromatography paths, perform the purification in a staggered and continuous manner, automatically collecting the desired fractions.

Interchim's flash/prep LC chromatography system puriFlash® 5.250 is ideally suited for this purpose.

Application Showcase

Method

Instrumental Set-up

A flow reactor system is used to perform synthesis and deliver the crude reaction mixture directly or via in-line extractors to the injection port of Advion Interchim Scientific's puriFlash® 5.250 chromatograph. The latter contains a 10-port valve that directs the flow of crude material alternately into one of two sample loops.

A repeating sequence, in which one of two identical flash columns is loaded with the reaction mixture while a chromatography run plus re-equilibration is performed on the other column, ensures that the contents of the sample loops are processed without substance loss. The separation is monitored using UV + ELSD detection and the fractions of interest are collected accordingly. (Figure 1)

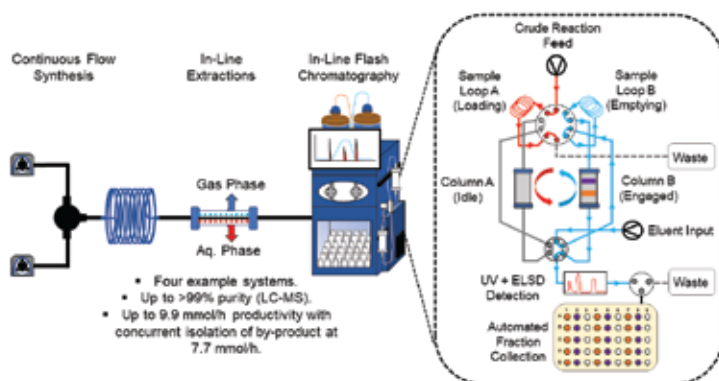
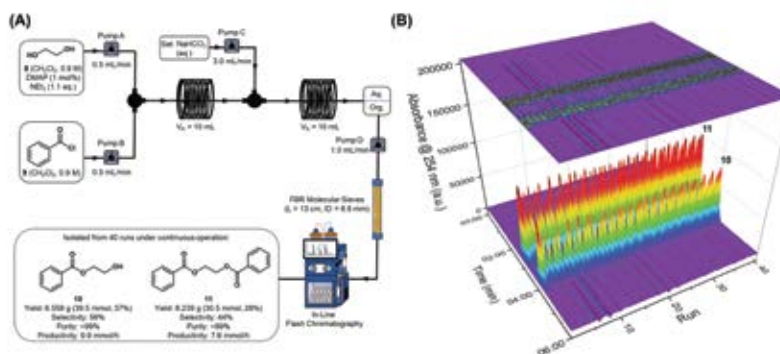


Figure 1: Typical instrumental setup

Figure 2: (A) Schematic flow diagram for the continuous flow esterification of ethylene glycol and benzoyl chloride with in-line normal-phase flash chromatography. DMAP: 4-(Dimethylamino) pyridine, FBR: Fixed Bed Reactor. (B) Stacked plot of 40 processed chromatograms from the continuous isolation of the ester products.



Principle of Method Development

The following systematic approach to achieve continuous purification is followed:

- » Optimization of flow reactor setup to obtain the maximum yield of products.
- » Minimizing the time required for flash chromatography-based purification of the crude reaction mixture in offline experiments.
- » Synchronizing the duration of the chromatography method with a suitable feed flow rate from the flow reactor to achieve a successful coupling.

Results

A proof-of-principle study using a test mixture (4-methoxyphenol and 2,5-dibromo-p-xylene in n-hexane/ethyl acetate) was successfully performed, proving the feasibility of flow chemistry - puriFlash® 5.250 integration and confirming the durability of Advion Interchim Scientific's flash column cartridges.

Encouraged by the success of the methodology, three different flow chemistry syntheses were successfully complemented with in-line normal or reversed-phase flash chromatography, resulting in the continuous isolation of products with 97-99% purity. It was realized that the flash chromatograph could also serve important secondary functions:

- » The possibility of cost-saving inline recovery of homogeneous (photo)catalysts was demonstrated using the example of the continuous flow oxidation of Fmoc-L-methionine with air to the corresponding sulfoxide via 4,7-diphenyl-2,1,3-benzothiadiazole as a homogeneous photosensitizer.
- » In-line process monitoring to ensure process consistency of a flow chemistry system could be conveniently performed using onboard flash chromatography. As an example, the quality control of continuous heterogeneous catalytic systems was

realized by inline reversed-phase flash chromatography for a flow coupling of thiophenol to diphenyl disulfide via heterogeneous photo-redox catalysis by TiO_2 nanoparticles.

- » The potential of the flow-integrated puriFlash® 5.250 to simultaneously isolate two products in grams per hour with >99% purity from a crude flow chemistry reaction mixture with polyfunctional substrates is described. In the continuous flow esterification of ethylene glycol and benzoyl chloride, productivity values of 9.9 and 7.6 mmol/h were achieved for the two desired esters.

Conclusion

Integrating our off-the-shelf flash chromatography system into your custom flow chemistry has no technological barrier to entry. The coupling easily adapts to the needs of the chemist and will help to intensify and accelerate chemical discovery. Automating routine flow synthesis of key intermediates required for compound library generation will particularly profit.

Reference

1. Thomson, C.; Banks, C.; Allen, M. G.; Barker, G.; Coxon, C. R.; Lee, A. -L.; Vilela, F. Expanding the Tool Kit of Automated Flow Synthesis: Development of In-Line Flash Chromatography Purification. *Journal of Organic Chemistry* 2021, 86 (20), 14079-14094. DOI: 10.1021/acs-joc.1c01151.


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